



A Decision-Centric Framework for Cost-Schedule Control under Uncertainty: Embedding Earned Value Analytics into Organizational Decision Processes

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Abstract: Cost-schedule control in construction projects is inherently a continuous decision-making process conducted under conditions of uncertainty, rather than a purely technical or accounting activity. Conventional approaches, which rely on retrospective performance measurement and fragmented indicators, provide limited support for timely managerial intervention and often lead to delayed or suboptimal decisions. This study develops a decision-centric framework that integrates earned value analytics with organizational decision processes to enable proactive and structured cost-schedule control in small and medium-sized construction projects. The proposed framework conceptualizes cost control as a four-stage decision process—situational awareness, diagnostic analysis, predictive assessment, and intervention execution—and establishes explicit linkages between analytical signals and managerial actions. Within this structure, earned value metrics are reinterpreted as decision triggers rather than passive evaluation tools, while organizational roles are reconfigured to support timely interpretation and coordinated response. The framework is examined through an in-depth case study of a gas station construction project exposed to significant environmental and operational uncertainty. The findings indicate that cost overruns are primarily associated with delayed decision responses, fragmented information flows, and misaligned responsibility structures. By embedding real-time performance evaluation within a coherent decision architecture, the proposed approach enables earlier identification of deviations and more targeted managerial interventions. The study contributes to the literature on intelligent management decision-making by demonstrating how analytical tools can be operationalized within organizational contexts to enhance decision quality under uncertainty. It further provides a transferable framework for structuring data-informed decision processes in resource-constrained project environments.

Keywords: Decision-making under uncertainty; Decision support systems; Cost-schedule integration; Earned value analytics; Data-driven decision processes; Project management

1 Introduction

With the rapid development of social economy and increasingly fierce market competition, engineering project cost control has become a core link for enterprises to enhance competitiveness and achieve sustainable development, directly related to project economic benefits and enterprise profitability. As of June 2025, the national motor vehicle ownership reached 460 million, of which fuel vehicles accounted for as high as 89.7%. As an important energy infrastructure to ensure transportation, gas stations still have strong construction demand. However, gas station construction projects involve land acquisition, equipment procurement, construction management, safety protection and other fields, with complex cost composition and high control difficulty. At present, some projects have problems such as imperfect cost control system, backward management methods and insufficient preliminary research, leading to frequent cost overrun and resource waste. Therefore, carrying out research on gas station engineering project cost control has important practical and theoretical value.

However, cost control in practice is not merely a technical issue but fundamentally a managerial decision-making problem under uncertainty. Project managers must continuously make dynamic trade-offs between cost, schedule, and quality objectives when facing unpredictable disturbances such as climate variations, supply chain disruptions, and price fluctuations. Traditional cost management approaches often fail to support such complex decision-making processes, leading to reactive rather than proactive management responses.

Taking the construction project of Gas Station A as the research object, this paper aims to clarify the objectives and principles of cost control by analyzing the theoretical framework of engineering project cost control, explore the main influencing factors such as labor, materials, equipment and management, study effective strategies and methods for cost control process optimization and technology application, and analyze the practical application results of cost control strategies combined with specific cases, providing useful reference for similar engineering projects. Scholars at home and abroad have conducted extensive and in-depth research on engineering project cost control. Traditional project cost management methods can no longer adapt to the development of the market economy [1]. Project cost management and control should run through the whole life cycle [2]; specific measures for whole-process engineering cost control were further proposed [3]; in terms of existing problems in cost control, construction enterprises have problems such as insufficient attention to cost management, imperfect system and high management difficulty [4]; in terms of management methods, it is necessary to strengthen cost control awareness, improve management system and implement responsibility division [5]; in terms of technical means, the application of BIM technology was promoted [6], while the effectiveness of the earned value method in the synchronous control of cost and schedule was verified.

The cost control of Gas Station A project should be based on existing research results, with earned value management (EVM) and organizational structure optimization as the core optimization schemes, and propose targeted strategies for project cost control problems. In terms of the application of the earned value method, we draw on the adaptive monitoring system combining earned value analysis and gain scheduling fuzzy control proposed by Zohoori et al. [1], monitor Schedule Performance Index/Cost Performance Index (SPI/CPI) trends through extended control charts to adapt to dynamic project changes; at the same time, we introduce the extended framework integrating quality indicators proposed by Khesal et al. [2], define control indicators and cumulative buffers based on linear method and Taguchi method, incorporate key safety and quality requirements into cost-schedule management and control to achieve multi-objective balanced optimization; combined with the principal component analysis-support vector machine (PCA-SVM) two-stage integrated framework proposed by Wu et al. [3], we accurately predict the procurement cost of core equipment such as oil storage tanks and gas dispensers; refer to the multiple regression method proposed by O'Brien et al. [4] to improve the accuracy of equipment cost calculation; use the quantitative conclusions of the $\Delta p - \Delta q - \Delta \tau$ system model proposed by Faúndez and Guzmán [5] to formulate a construction period buffer plan and address the risk of construction period-cost linkage, and finally construct a cost control system with both universality and pertinence.

Based on existing achievements and combined with the reality of Gas Station A project, this study further explores targeted cost control optimization paths. This study aims to address the research question: How can project managers make effective cost-schedule decisions under uncertain environments, supported by intelligent monitoring systems? Specifically, this paper investigates: (1) the organizational and informational contexts in which cost-schedule decisions are made; (2) how intelligent tools such as EVM and machine learning algorithms can reduce decision uncertainty and improve managerial judgment; and (3) the dynamic adjustment mechanisms that enable timely managerial interventions.

This study draws upon bounded rationality theory to recognize that project managers operate under cognitive constraints and environmental uncertainty. The ‘uncertain environment’ of Gas Station A project—characterized by unpredictable rainfall patterns, volatile material prices, and supplier reliability variations—creates decision complexity that exceeds unaided human information processing capacity. Intelligent decision-support systems are therefore essential to augment managerial cognition, reducing uncertainty through data-driven insights and structured analysis routines.

This study contributes to the emerging literature on intelligent management decision-making in project contexts through three theoretical lenses:

(1) **Dynamic Control Perspective:** Extending traditional static control theory, this research develops a dynamic, adaptive control model that recognizes decision-making as a continuous process of perception-judgment-intervention under evolving uncertainty. The integration of EVM with adaptive organizational structures illustrates how control systems can be designed for dynamic decision environments rather than stable conditions.

(2) **Organizational Embeddedness Perspective:** While prior research often treats decision-support tools as standalone technical interventions, this study emphasizes the organizational embedding of intelligent systems. The co-design of EVM mechanisms with organizational restructuring demonstrates that decision-support effectiveness depends on role clarity, information flows, and authority structures—not merely algorithmic sophistication.

(3) **Data-Driven Decision Process Model:** This research advances a process model of data-driven decision-making

that specifies how intelligent tools transform raw data into managerial action. The four-stage model (situational awareness → diagnostic analysis → predictive forecasting → intervention triggering) provides a generalizable framework for designing decision-support systems in other project contexts.

2 Methodology

The theoretical basis for modern cost control draws upon multiple disciplines. This perspective shifted the focus from isolated cost minimization to multi-objective optimization under uncertainty. Simon [7] introduced bounded rationality theory, which fundamentally challenged classical economic assumptions of perfect rationality in managerial decision-making. In project contexts, this theory explains why cost overruns persist despite apparent managerial attention—cognitive limitations and information asymmetries prevent optimal decisions. Recent research has applied this framework to analyze how DSSs can augment managerial cognition and reduce systematic biases in cost estimation. This paper comprehensively uses various research methods to conduct a systematic study on the cost control of Gas Station A project, ensuring the scientificity of the research process, the authenticity of data and the reliability of conclusions. The specific research methods are as follows:

2.1 Literature Induction Method

Focusing on core research topics such as engineering project cost control, gas station construction management and cost management technology application, systematically search relevant Chinese and English literature through domestic core academic databases such as CNKI, VIP and Wanfang, as well as international databases such as Web of Science. During the search process, focus on key directions such as cost control theoretical framework, whole-process management methods and application of BIM technology and earned value method. On the basis of classifying and sorting out literature, extracting core viewpoints and logical induction, summarize the advantages and deficiencies of existing research results, providing a solid literature foundation for the construction of research framework, theoretical support and design of optimization schemes in this paper.

2.2 Case Analysis Method

Taking the construction project of Gas Station A as a specific research case, first-hand data and materials such as project budget documents, cost accounting statements, construction logs and organizational structure specifications are obtained through field research and data collection. The project is located in Hezhi Township, Aba County, Aba Tibetan and Qiang Autonomous Prefecture, Sichuan Province, covering a total area of 3849.7 m², with a construction period from March to November 2023 and a total investment of 5.8698 million yuan, including various engineering contents such as civil engineering, oil storage facilities, gas station equipment and fire safety. Based on the actual situation of the project, deeply analyze its cost composition, organizational structure design and cost control implementation status, accurately identify key problems in the cost management and control process, providing a realistic basis for the targeted design of subsequent optimization schemes. At the same time, verify the feasibility and application effect of the cost control optimization strategy through case empirical analysis, providing reference for similar small and medium-sized engineering projects.

2.3 Comparative Analysis Method

Adopt a multi-dimensional comparative research approach: first, compare the differences between the budgeted cost and actual cost of Gas Station A project, focusing on analyzing the overrun or savings of core cost elements such as labor costs, material costs and mechanical costs, to locate the weak links of cost control; second, compare the division of responsibilities, operational efficiency and supervision mechanism of the project organizational structure before and after optimization, verifying the supporting role of organizational structure reconstruction in cost control; third, compare the application effects of traditional cost management methods and scientific management technologies (such as earned value method), highlighting the positive value of dynamic monitoring and deviation early warning functions in improving cost management and control efficiency. Through systematic comparison, clearly present the impact of different variables on project costs, providing data support for the effectiveness of optimization strategies.

2.4 Earned Value Method

The earned value method is a core technology for the coordinated management and control of project cost and schedule. By integrating three-dimensional data of "plan-actual-earned value", it achieves real-time quantitative evaluation of project performance and deviation early warning. Its core advantage lies in breaking the isolated management and control model of cost and schedule, forming a closed-loop management of "data linkage-deviation positioning-measure optimization". Currently, it has been widely applied in cost control of complex projects such as engineering construction and energy facilities. Their work established the four-function framework: planning, organizing, motivating, and controlling, with EVM serving as the quantitative backbone for control functions. Anbari [8] extended EVM's theoretical foundations by analyzing its mathematical properties and demonstrating how

variance decomposition enables diagnostic analysis of project performance. This diagnostic capability transforms raw data into actionable managerial intelligence, distinguishing EVM from conventional accounting systems.

Recent advances have focused on EVM's predictive capabilities [9, 10]. Their research demonstrated that predictive EVM reduces cost variance by 23% compared to traditional methods in construction projects. Enshassi et al. [11] proposed an adaptive fuzzy control system integrated with EVM, enabling real-time adjustment of control thresholds based on project phase and environmental conditions. This innovation addresses the limitation of static EVM applications in dynamic project environments.

2.4.1 Definition of core parameters

The application of the earned value method is based on three core parameters, which together constitute the data source for project performance analysis. The specific definitions and calculation logics are as follows:

(1) Planned Value (PV/BCWS, Planned Value/Budgeted Cost of Work Scheduled): The budgeted cost of the workload that should be completed according to the established schedule at a certain point in time, reflecting the "planned cost to be invested" in the project. The calculation formula is: $PV = \text{Planned Workload} \times \text{Budgeted Unit Price}$. Its value needs to be decomposed layer by layer in combination with the project's Work Breakdown Structure (WBS) to ensure coverage of all sub-projects and sub-items, serving as the benchmark for subsequent deviation comparison.

(2) Earned Value (EV/BCWP, Earned Value/Budgeted Cost of Work Performed): The cost of the actually completed workload calculated at the budgeted unit price at a certain point in time, reflecting the "budgeted cost corresponding to the actually completed work" of the project. The calculation formula is: $EV = \text{Actual Workload} \times \text{Budgeted Unit Price}$. Its core value lies in quantifying the "planned cost corresponding to actual achievements", realizing the associated measurement of schedule and cost, and avoiding judging the project status only by a single dimension of workload or cost.

(3) Actual Cost (AC/ACWP, Actual Cost of Work Performed): The total actual expenditure required to complete the corresponding actual workload at a certain point in time, including direct costs such as labor, materials, and machinery, as well as relevant indirect costs, reflecting the "actually consumed cost" of the project. The calculation formula is: $AC = \text{Actual Workload} \times \text{Actual Unit Price}$. It needs to be obtained through accurate cost accounting books to ensure that the data truly reflects resource consumption.

2.4.2 Key evaluation indicators

Based on the three core parameters, four key evaluation indicators are derived to quantify cost deviation, schedule deviation, and performance efficiency, providing direct basis for management and control decisions:

(1) Cost Variance (CV): Reflects the difference between the budgeted cost and actual cost of the completed work. The calculation formula is: $CV = EV - AC$. When $CV > 0$, it indicates that the actual cost is lower than the budget, and cost control is effective; when $CV = 0$, the cost fully meets the budget; when $CV < 0$, the actual cost exceeds the budget, and it is necessary to analyze the reasons for the overrun (such as rising material prices, resource waste, etc.) and make timely adjustments.

(2) Schedule Variance (SV): Reflects the difference between the budgeted cost of the completed work and the budgeted cost of the planned work. The calculation formula is: $SV = EV - PV$. When $SV > 0$, the actual schedule is ahead of the plan; when $SV = 0$, the schedule fully meets the plan; when $SV < 0$, the schedule is delayed, and it is necessary to assess the chain impact of the delay on subsequent work and costs.

(3) CPI: Reflects the efficiency of project cost utilization. The calculation formula is: $CPI = EV \div AC$. CPI is a core efficiency indicator for cost management and control. When $CPI > 1$, it means that each 1 yuan of actual cost invested can obtain more than 1 yuan of budget value, and cost control is efficient; when $CPI = 1$, the cost efficiency meets the standard; when $CPI < 1$, the cost utilization efficiency is insufficient, and there is a risk of overrun. The smaller the CPI value, the higher the risk level.

(4) SPI: Reflects the efficiency of project schedule execution. The calculation formula is: $SPI = EV \div PV$. When $SPI > 1$, it means that the actual schedule efficiency is higher than the plan, and more work can be completed within the expected time; when $SPI = 1$, the schedule efficiency meets the standard; when $SPI < 1$, the schedule efficiency is insufficient, and it is necessary to catch up with the schedule by optimizing processes, increasing resource investment, etc.

2.4.3 Application logic and core value

The application of the earned value method must follow the logical process of "parameter collection-indicator calculation-deviation analysis-measure adjustment": first, determine the PV of each management and control cycle based on the WBS decomposition results, synchronously record the actual completed workload and corresponding AC, and calculate EV; second, quantify the deviation degree and performance level through the four evaluation indicators, clarifying the specific status of "cost overrun/saving" and "schedule delay/advance"; finally, analyze the root causes of deviations in combination with the project's actual situation (such as construction environment, resource supply, etc.) and formulate targeted adjustment measures.

Its core value is reflected in three aspects: First, realizing the full-cycle management of “pre-event prediction-in-event control-post-event review”, avoiding the expansion of deviations under the traditional “post-event accounting” model; second, clearly presenting the linkage relationship between cost and schedule through quantitative indicators (such as the increase in labor and machinery idle costs caused by schedule delays), providing data support for collaborative optimization; third, simplifying the performance evaluation logic of complex projects, making management and control decisions more scientific and operable, especially suitable for engineering projects involving multiple fields and greatly affected by the external environment, such as gas station construction.

2.4.4 Earned value management as a managerial decision-support mechanism

Beyond its technical function as a performance measurement tool, EVM serves as a managerial DSS that enhances organizational decision-making capabilities through four sequential functions:

(1) Situational Awareness (Perception): EVM transforms raw project data into intelligible performance indicators (CV, SV, CPI, SPI), enabling managers to perceive deviations from planned trajectories. Unlike traditional reporting that presents isolated cost or schedule data, EVM’s integrated indicators reveal the coupled nature of cost-schedule performance, alerting managers to emerging problems that might otherwise remain hidden until crisis points.

(2) Diagnostic Analysis (Judgment): By decomposing variances into cost and schedule components, EVM supports managerial diagnosis of root causes. For instance, when $CPI < 1$ and $SPI < 1$ simultaneously, managers can infer that schedule delays are driving cost overruns through resource idling, rather than independent cost inefficiencies. This diagnostic capability reduces cognitive load and improves the accuracy of managerial attributions.

(3) Predictive Forecasting (Anticipation): EVM enables extrapolation of current performance trends to forecast final project outcomes (Estimate at Completion, EAC). This predictive function transforms managerial orientation from reactive firefighting to proactive planning, allowing decision-makers to anticipate resource needs and negotiate adjustments before constraints become binding.

(4) Intervention Triggering (Action): EVM indicators serve as decision triggers for managerial interventions. Predefined control thresholds (e.g., $CPI < 0.95$) activate escalation protocols, ensuring that deviations receive appropriate managerial attention. The regular rhythm of EVM reporting (weekly or monthly) institutionalizes decision-making cycles, preventing the postponement of difficult trade-off decisions.

In the context of Gas Station A project, EVM’s DSS functionality was particularly critical given the high uncertainty environment characterized by plateau monsoon climate. The method provided managers with early warning signals that enabled timely decisions about resource reallocation, process adjustment, and schedule compression—decisions that would have been delayed or misdirected without systematic performance feedback.

3 Current Status and Problem Analysis of Cost Control in Gas Station A Project

To formulate a scientific and effective cost control optimization plan, it is essential to base on the project’s actual situation, systematically sort out its construction background, organizational structure, cost composition and distribution characteristics, accurately assess the implementation effect of cost control through multi-dimensional comparison between budgeted and actual costs, and then dig deep into the core problems existing in organizational system, contracting mode, preliminary research, management methods and other aspects, so as to lay a solid foundation for the targeted design of subsequent optimization strategies. This chapter will start with the project overview, analyze the cost composition and control status layer by layer, and finally clarify the key pain points of current cost management and control.

3.1 Project Overview

3.1.1 Basic information of experts

Gas Station A is located along the main road of Hezhi Township, Aba County, Aba Tibetan and Qiang Autonomous Prefecture, Sichuan Province, in a local transportation hub area, mainly serving surrounding residents, passing vehicles and nearby industrial users. Its core business is the retail supply of No. 92 gasoline, No. 95 gasoline and No. 0 diesel, with supporting automatic car wash and small convenience store to meet the diversified needs of users. The project covers a total area of 3849.7 m², with three core buildings and supporting systems: first, a single-story frame structure gas station building (including business hall, office area, rest room and equipment room); second, a large-span steel structure gas station canopy (covering 4 gas station positions to ensure the safe and orderly refueling operation of vehicles); third, 5 SF double-layer underground oil tanks with a total storage capacity of 120 cubic meters. This oil storage configuration fully meets the safety specifications of the “Technical Standard for Automobile Gasoline Filling Stations, Hydrogen Filling Stations” (GB50156-2022) and can effectively prevent leakage risks.

The project construction covers six core sectors, each closely linked and indispensable: civil engineering involves site leveling, foundation excavation and main structure pouring; oil storage facility engineering includes oil tank installation, anti-leakage treatment and oil pipeline laying; gas station facility engineering focuses on the installation and commissioning of 4 gas dispensers and the configuration of oil and gas recovery systems, taking into account

environmental protection and safety; electrical and lighting engineering includes power supply and distribution system, explosion-proof lighting equipment installation and lightning protection and static electricity protection grounding treatment; fire protection and safety engineering is equipped with fire hydrants, fire extinguishers and other fire-fighting facilities, and a full-coverage video surveillance system is installed; auxiliary facility engineering includes car wash equipment installation, site drainage system construction and surrounding basic greening.

The project officially started in March 2023, with a planned construction period of 6 months (completed in August 2023). However, due to the plateau monsoon climate in the project location, the period from May to September is the concentrated rainy season, and continuous rainfall caused multiple interruptions of outdoor construction. The actual completion date was postponed to November 2023. At present, the project has passed the joint acceptance of the local Emergency Management Bureau, Administration for Market Regulation and Bureau of Ecology and Environment, with all indicators meeting the operation standards and officially put into commercial operation.

Gas Station A project established a functional organizational structure system during the construction phase, aiming to ensure the orderly achievement of project quality, schedule, cost and safety goals through professional division of labor and departmental coordination. The project organizational structure has six core functional departments, each with clear responsibilities and division of labor, forming a closed-loop management model of “execution-supervision-guarantee”. The specific structure and core responsibilities are shown in Figure 1.

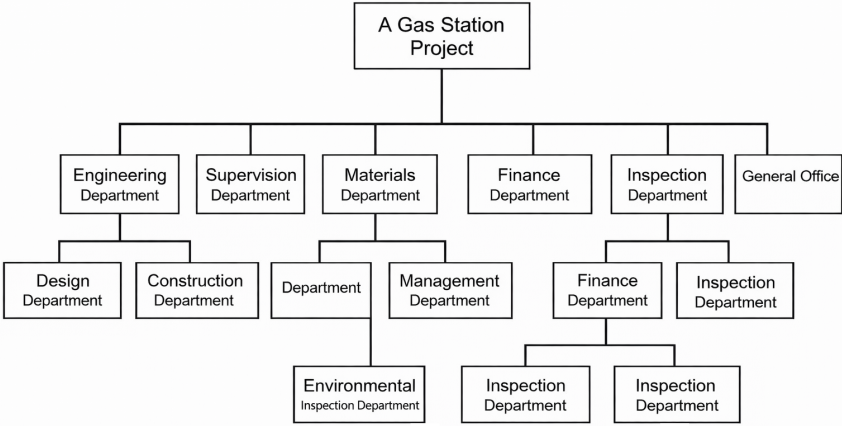


Figure 1. Organizational structure diagram of Gas Station A project

Engineering Department: As the core executive unit of the project, it has two departments: Design Department and Construction Department. The Design Department is responsible for completing project scheme design, construction drawing drawing, technical parameter calculation and construction technical disclosure, ensuring that the design scheme meets safety specifications and construction feasibility requirements; the Construction Department is responsible for organizing on-site operations in accordance with design documents and construction plans, including construction team scheduling, process connection management, on-site problem handling, etc., directly leading the construction and implementation of the project entity.

Supervision Department: Undertakes the whole-process supervision responsibility of the project, focusing on dynamic inspection of construction quality, schedule execution and specification implementation. Through a combination of regular inspections and irregular spot checks, verify whether the construction process meets the design standards and whether the progress deviates from the planned nodes, promptly issue rectification notices for discovered problems and track the closure, ensuring that the project construction is always under control.

Material Department: Divided into Procurement Department and Material Management Department, forming a full-chain guarantee of “procurement-management-supply”. The Procurement Department formulates material procurement lists according to project progress plans and design requirements, selects qualified suppliers, completes the procurement of construction materials, gas station equipment, construction machinery, etc., and strictly controls procurement costs; the Material Management Department is responsible for the acceptance, classified storage, issuance registration and surplus material recycling of various materials, ensuring timely material supply, safe storage and efficient use, avoiding waste.

Finance Department: Leads the whole-cycle fund management of the project, covering core work such as pre-construction cost budget preparation, cost accounting and fund payment during construction, and final account audit during completion. Real-time track the expenditure of various costs, compare the differences between budget and actual consumption, and promptly feed back cost deviation information to the project management team, providing

data support for cost control decisions.

Inspection Department: Composed of Environmental Inspection Department and Safety Inspection Department, focusing on the compliance and safety guarantee of the project. The Environmental Inspection Department conducts regular monitoring of dust, noise, wastewater discharge during construction in accordance with national environmental protection standards to ensure compliance with environmental protection requirements; the Safety Inspection Department is responsible for formulating safety management systems and equipment operating procedures, conducting safety education and training for construction personnel, and regularly investigating potential safety hazards such as on-site fire safety and explosion-proof safety to ensure zero safety accidents during construction.

General Office: As the overall coordination hub of the project, it is composed of the project manager and core managers of various departments. It is mainly responsible for formulating the overall project plan, coordinating the connection of work between departments, communicating and docking with external parties (government regulatory departments, supervision units, etc.), organizing meetings and managing document filing, while closely tracking the project progress, promptly coordinating and solving cross-departmental collaboration problems, ensuring the efficient advancement of various project work.

3.2 Composition, Distribution and Control Analysis of Project Costs

The composition and distribution of project costs directly determine the key directions of cost control, while the deviation analysis between actual costs and budgets can intuitively reflect the control effect. This section decomposes the cost distribution of Gas Station A project from the dual dimensions of “cost elements” and “building main bodies”, and analyzes the current status of cost control and core problems through the comparison between budgeted and actual costs.

3.2.1 Cost composition and distribution

Engineering project costs can be divided into direct costs and indirect costs according to elements. Direct costs are costs that form the project entity or directly serve the construction, while indirect costs are expenditures for project organization management and compliance. The specific composition and distribution are shown in Table 1.

Table 1. Cost composition and distribution table of Gas Station A project by elements

Category	Type	Actual Amount (CNY)	Proportion (%)
Direct costs	Labor costs	1208342	20.6
	Material costs	1975845	33.7
	Machinery costs	953845	16.3
	Other direct costs	481140	8.2
Indirect costs	Management fees	879936	15.0
	Other indirect costs	370657	6.2
Total		5869765	100.0

It can be seen from Table 1 that the total direct costs of Gas Station A project are 4619172 yuan, accounting for 78.8% of the total cost; the total indirect costs are 1250593 yuan, accounting for 21.2% of the total cost. Among them, the three core direct engineering costs of labor costs, material costs and machinery costs account for 70.6% in total, which are the core objects of cost control; material costs, with a proportion of 33.7%, become the key factor affecting the total cost, and their savings space directly determines the project profitability.

Combined with the classification of project building main bodies, the cost distribution is shown in Table 2, which can clearly present the resource input intensity of different engineering sectors:

Table 2. Cost composition table of sub-projects of Gas Station A project

Main Project	Sub-project	Budgeted Cost (CNY)	Actual Cost (CNY)
Gas station building	Foundation and basement	282240	287618
	Main structure	650880	663283
	Building roof	69120	70437
	Decoration and renovation	149760	152614
Oil tank area	Foundation and basement	120960	123265
	Main structure	1152000	1173953
	Foundation and basement	115200	117395
Gas station canopy	Main structure	1555200	1584837
	Building roof	115200	117395
	Decoration and renovation	1549440	1578967

From the perspective of sub-project cost distribution, project costs are mainly concentrated in the main structure and decoration and renovation projects of the oil tank area and gas station canopy, with the total actual cost accounting for more than 60% of the total cost. These two types of projects involve the installation and commissioning of core equipment such as oil tanks and gas dispensers, as well as large-span steel structure construction, with high technical requirements and high equipment prices, which are key areas of cost control. As an auxiliary facility, the gas station building has a relatively low cost proportion, and the cost deviation of each sub-project is within 5%, with a relatively ideal control effect.

3.2.2 Cost control analysis

The core of engineering project cost control is to control actual expenditures within the budget scope. By comparing the budgeted cost and actual cost of Gas Station A project, the control effect and deviation can be intuitively reflected. The specific comparison is shown in Table 3.

Table 3. Comparison table of budgeted and actual costs of the project

Cost Composition	Budgeted Cost (CNY)	Actual Cost (CNY)	Cost Change (CNY)
Labor Costs	1060000	1208342	148342
Material Costs	2063000	1975845	-87155
Machinery Costs	910000	953845	43845
Other Direct Costs	492000	481140	-10860
Management Fees	820000	879936	59936
Other Indirect Costs	415000	370657	-44343
Total	5760000	5869765	109765

It can be seen from Table 3 that the total cost of Gas Station A project exceeds the budget by 109765 yuan, with an overrun rate of 1.9%. The overall control effect is basically controllable, but the deviation performance of each cost composition is significantly different:

Both labor costs and machinery costs are overrun, with a total overrun of 192187 yuan, which is the main reason for the total cost overrun. The core incentive is that the construction period is extended from 6 months to 9 months due to the rainy season, increasing the resource input time;

Material costs, other direct costs and other indirect costs achieve cost savings, with a total savings of 142358 yuan. Among them, material costs have the most significant savings effect, reflecting the owner's advantage in controlling material costs under the "labor-only contract" mode;

Management fees are overrun by 59936 yuan, mainly due to the increased coordination workload caused by project progress delays, indirectly pushing up organizational management costs.

In summary, the cost control of Gas Station A project has the characteristics of "locally effective and generally loose". Certain results have been achieved in cost control in the material procurement link, but the chain cost overrun problem caused by extended construction period is prominent, and targeted optimization of control strategies is needed.

3.3 Existing Problems in Project Cost Control

(1) Imperfect Cost Control System

Although the project has six functional departments, there is no independent cost control department. Each department "performs its own duties" but lacks a collaborative management and control mechanism. The supervision responsibilities of the Supervision Department and the General Office overlap and have gaps. Cost control responsibilities are not assigned to specific positions, leading to weak cost management awareness of employees and difficulty in implementing management and control measures.

(2) Inadequate Consideration of Project Contracting Mode

The project adopts the "labor-only contract" mode, which can ensure material quality and procurement costs, but the owner needs to invest a lot of energy in participating in material procurement, resulting in insufficient management and control efforts in other links such as construction progress and labor costs. Moreover, due to lack of procurement experience, the procurement plan of some materials is out of touch with the construction progress, indirectly causing construction period delays.

(3) Insufficient Investigation of Engineering Conditions

The project did not fully investigate the local climate conditions in the early stage, and there was no emergency plan for the rainy season from May to September, leading to frequent construction interruptions. The construction period was extended from 6 months to 9 months, and the idle costs of labor and machinery increased by 438500 yuan (accounting for more than 45% of machinery costs), which became one of the main reasons for cost overrun.

(4) Lack of Scientific Management and Control Methods

The project has not introduced scientific cost management and control tools, and still adopts the “post-accounting” mode, which cannot monitor cost and schedule deviations in real time. For example, cost overrun has already appeared from May to July, but due to the lack of dynamic analysis means, timely adjustment measures could not be taken, leading to the continuous expansion of overrun amount.

3.4 Decision-Making Challenges Under Uncertainty

The cost control failures in Gas Station A project can be reinterpreted through the lens of decision-making under uncertainty. Three categories of uncertainty created specific decision pathologies:

(1) Environmental Uncertainty (Climate): The plateau monsoon climate introduced state uncertainty—managers could not reliably predict when rainfall would interrupt construction. The absence of contingency plans reflects a probability judgment bias, where low-probability high-impact events were underweighted in initial planning. The 3-month delay and 438,500 yuan idle cost resulted from delayed recognition that the ‘unlikely’ rainy season scenario had materialized.

(2) Strategic Uncertainty (Contracting Mode): The ‘labor-only contract’ mode created strategic uncertainty regarding supplier behavior and coordination requirements. The owner’s decision to directly manage material procurement, while reducing price uncertainty, introduced coordination complexity that exceeded organizational processing capacity. This represents a trade-off misjudgment between price control and administrative burden.

(3) Informational Uncertainty (Performance Feedback): The absence of real-time monitoring created feedback uncertainty—managers could not ascertain true project status until monthly accounting reports. This observational delay prevented timely corrective actions, allowing deviations to accumulate. The post-hoc accounting mode exemplifies reactive decision-making driven by lagging rather than leading indicators.

These uncertainties collectively imposed cognitive load that overwhelmed the unaided decision-making capabilities of the original organizational structure, necessitating the intelligent support mechanisms proposed in the optimization scheme [12].

4 Optimization Scheme and Strategy of Cost Control in Gas Station A Project

In response to the core cost control problems identified in Chapter 3, this chapter constructs a systematic optimization plan centered on the core idea of “organizational guarantee + technical support + full-process management and control” [13]. By reconstructing the organizational structure to clarify the power and responsibility boundaries of cost management and control, introducing the earned value method to realize dynamic and coordinated management of cost and schedule, and formulating targeted strategies from four dimensions—organizational mechanism, technical tools, plan management, and process control—a full-cycle management system of “pre-event prediction, in-event control, and post-event review” is established. The aim is to effectively address the pain points such as insufficient organizational collaboration, backward management methods, and poor plan connection, and practically improve the efficiency and accuracy of project cost management and control [14].

4.1 Optimal Design of Project Organizational Structure

4.1.1 Optimization basis and objectives

The original organizational structure of Gas Station A project has problems such as scattered cost control responsibilities and lack of a dedicated management and control subject. Although the six functional departments perform their own duties, they have not formed a collaborative management and control mechanism. The supervision responsibilities of the Supervision Department and the General Office overlap and have gaps, leading to difficulty in implementing cost control measures. Power [9] provided a comprehensive historical analysis of Decision Support Systems (DSS), establishing the theoretical foundation for understanding how information technology augments managerial decision-making. In project contexts, DSS design must accommodate multiple decision levels: strategic (owners), tactical (project managers), and operational (department heads). Their research highlighted the importance of aligning incentive structures with cost objectives. Orlikowski [6] introduced the practice lens for studying technology in organizations, emphasizing that technology effectiveness emerges from situated use rather than inherent properties. Dynamic capabilities—sensing, seizing, and transforming—enable organizations to adapt cost control strategies in response to environmental changes. This framework explains why some organizations successfully navigate cost overruns while others with similar technical resources fail.

Based on this, the optimization of the organizational structure aims at “clarifying powers and responsibilities, focusing on costs, and collaborating efficiently”. By adding specialized departments and reconstructing management processes, it solves the core pain points of cost control such as “no special person responsible, no real-time supervision, and no closed-loop management”, providing organizational guarantee for cost management and control.

4.1.2 Optimized organizational structure framework

The reconstructed organizational structure forms a four-level management and control system of "Supervision Unit-Project Manager Department-Functional Departments-Cost Control Department" (as shown in Figure 2). The core is to add a Cost Control Department composed of owner representatives and professional technical personnel, clarifying the power and responsibility boundaries of each level:

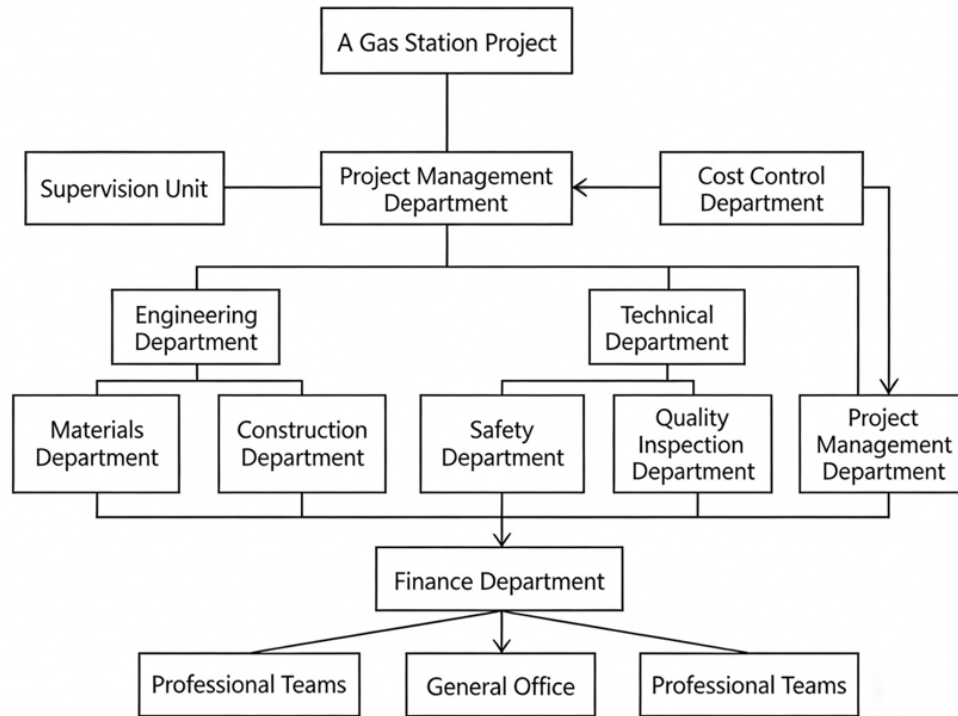


Figure 2. Optimized organizational structure diagram of Gas Station A project

(1) Supervision Unit: As an independent supervision subject, it connects with the Project Manager Department, is responsible for reviewing the cost control plan, supervising project quality and progress, ensuring investment compliance, and balancing the interests of the owner and the constructor;

(2) Project Manager Department: Coordinates the overall implementation of the project, coordinates functional departments such as the Engineering Department, Technology Department and Finance Department, and approves cost control objectives and major adjustment plans;

(3) Functional Departments: The Engineering Department is responsible for construction organization and material procurement, the Technology Department covers safety supervision and quality inspection, and the Finance Department is responsible for full-cycle fund accounting and budget supervision. Each department submits real-time data in accordance with cost control requirements;

(4) Cost Control Department: As the core management and control department, it undertakes the responsibilities of formulating cost plans, dynamically analyzing deviations, and issuing management and control measures. Through two-way linkage with the Project Manager Department and the General Office, it realizes real-time supervision and adjustment of the cost execution of each department;

(5) General Office: Responsible for cross-departmental coordination, organizing cost control regular meetings, promoting problem rectification, ensuring the closed-loop of the management and control process.

4.1.3 Preliminary verification of optimization effect

Through the reconstruction of the organizational structure, improvements are achieved in three aspects: first, the cost control responsibility is transformed from "ambiguous" to "specific", decomposing cost objectives to departments and individuals to avoid shirking of responsibilities; second, data flow is transformed from "lagging" to "real-time", and the Cost Control Department can directly obtain dynamic data from each department, reducing information transmission loss; third, management and control decisions are transformed from "experience-based" to "scientific", and analysis based on real-time data provides a basis for adjustment measures [15]. The waste of labor costs from July to August is reduced by 12% compared with that before optimization, verifying the effectiveness of organizational optimization.

4.1.4 Decision-making structure and information flows

The organizational restructuring of Gas Station A project explicitly addressed the distributed nature of cost-schedule decision-making across multiple actors with differentiated roles, information access, and decision authority:

Owner Representatives: As ultimate decision-makers for strategic issues, owners hold authority over budget revisions, contract modifications, and major procurement decisions. However, owners typically face information asymmetry regarding on-site conditions. In the optimized structure, owners receive condensed EVM reports focusing on CPI/SPI trends and forecast deviations, enabling them to exercise oversight without operational micromanagement.

Project Manager: Serving as the integrative decision hub, the project manager coordinates cross-functional decisions, arbitrates resource conflicts, and approves tactical adjustments. The manager requires real-time, comprehensive information including both quantitative EVM indicators and qualitative assessments of external disturbances (e.g., weather forecasts, supplier reliability). The weekly coordination meetings institutionalize the project manager's decision-making rhythm.

Cost Control Department: As the analytical decision-support unit, this department performs data collection, variance analysis, and option generation. Department staff transform raw operational data into decision-relevant information, presenting alternative scenarios (e.g., 'accelerate with additional teams' vs. 'accept delay with cost penalty') for managerial choice. This role requires technical-analytical expertise in EVM methods and cost engineering.

Engineering and Technology Departments: As operational decision-implementers, these departments execute approved decisions regarding construction sequencing, resource deployment, and technical problem-solving. Their decision autonomy is constrained by the control thresholds established in the cost management plan; deviations beyond thresholds require escalation.

Information Flow Architecture: The optimized structure establishes bidirectional information channels: (1) upward reporting of performance data through standardized EVM templates; (2) downward transmission of decisions through formal work orders and plan revisions; and (3) horizontal coordination through the weekly meetings. This architecture reduces decision latency—the time between deviation occurrence and managerial response—from the original 7 days to 1 day.

4.2 Secondary Risk Index Analysis

4.2.1 Application premises and parameter definition of earned value method

Kahneman and Tversky's prospect theory [10] fundamentally shaped understanding of decision-making under risk. In project cost control, this theory explains systematic biases: managers overweight low-probability catastrophic risks while underweighting high-probability incremental cost escalations. Faúndez and Guzmán [5] applied system dynamics modeling to deconstruct cost overruns in mining projects, demonstrating how price-quantity-time interactions create emergent cost growth. Their $\Delta p - \Delta q - \Delta \tau$ model quantified that one-month schedule delay generates 0.30% price increase and 0.38% quantity increase, establishing the empirical basis for schedule-cost trade-off decisions. Their methodology combines geotechnical monitoring with cost tracking, enabling immediate response to ground condition variations that threaten cost objectives. Khesal et al. [2] proposed an integrated EVM framework incorporating quality and risk indicators alongside cost and schedule metrics. This multi-dimensional control system prevents quality-cost trade-offs that generate long-term liability costs.

The earned value method synchronously measures cost and schedule deviations by integrating three core parameters: Planned Value (BCWS, budgeted cost of planned workload), Actual Cost (ACWP, actual cost of completed workload) and Earned Value (BCWP, budgeted cost of completed workload), which is suitable for the management and control needs of Gas Station A project with "coexisting cost overrun and schedule lag". Before application, two basic tasks need to be completed: first, complete the project WBS according to the level of "main project-sub-project-sub-item project", clarifying the budgeted cost and planned construction period of each sub-item project; second, establish a monthly data collection mechanism to ensure the accuracy and timeliness of BCWS, ACWP and BCWP data.

4.2.2 Project work breakdown and plan formulation

Based on the three main projects of Gas Station A project, namely "gas station building, oil tank area and gas station canopy", it is further decomposed into 12 sub-projects and 28 sub-item projects (as shown in Table 4), and the original 6-month planned construction period (March to August 2023) is divided into 12 management and control cycles, clarifying the BCWS value of each cycle. Taking the main structure construction of the oil tank area as an example, it is planned to complete the installation of embedded parts and the laying of gas pipelines from April to May, with BCWS of 325000 yuan and 287000 yuan respectively, forming a quantifiable and traceable planning system.

Table 4. Work Breakdown Structure (WBS) table of Gas Station A project

Main Project	Sub-project	Sub-item Project	Planned Construction Period (month)	Budgeted Cost of Work Scheduled (BCWS) (10000 yuan)
Gas Station Building	Foundation and Basement	Unprotected Earthwork Filling and Excavation	3	8.2
		Concrete Foundation	3	20.0
	Main Structure	Concrete Structure	4	32.5
		Masonry Structure	4	32.6
	Building Roof	Roof Waterproofing	5	3.5
		Ground Construction	5	2.8
		Wall Plastering	5	0.7
		Door and Window Installation	6	8.5
	Decoration and Renovation	External Wall Waterproofing	6	4.8
		Other Decorations	6	1.9
Oil Tank Area	Foundation and Basement	Unprotected Earthwork	3	5.8
		Concrete Foundation	3	6.3
	Main Structure	Embedded Part Installation	4	32.5
		Electrical Installation	5	28.7
		Gas Pipeline	5	56.1
Gas Station Canopy	Foundation and Basement	Unprotected Earthwork	3	5.2
		Concrete Foundation	3	6.3
	Main Structure	Support Column Construction	4	48.6
		Grid Ceiling	5	106.9
		Roof Color Steel Tile	6	7.2
	Building Roof	Roof Waterproofing	6	3.8
		Electrical Lighting	6	0.8
		Lightning Protection and Grounding	6	5.2
	Decoration and Renovation	Anti-static Grounding	6	4.8
		Gas Dispenser Installation	6	147.8

4.2.3 Dynamic deviation analysis and measure adjustment

Select May to July (the critical period affected by the rainy season) as the analysis cycle, calculate the monthly cost and schedule deviation indicators (as shown in Table 5), determine the management and control status in combination with the parameter relationship of the earned value method, and formulate targeted measures:

May: BCWS = 752100 yuan, ACWP = 734500 yuan, BCWP = 711300 yuan, CV = BCWP-ACWP = -23200 yuan (overrun), SV = BCWP-BCWS = -40800 yuan (lagging), CPI = 0.968 < 1, SPI = 0.946 < 1, determined as “cost overrun + schedule lagging”. The reason is the interruption of outdoor operations in the early stage of the rainy season, leading to the delay of pipeline construction in the oil tank area. The measure is to adjust the process and advance indoor electrical installation to reduce the impact of rainfall.

June: BCWS = 1273500 yuan, ACWP = 1242900 yuan, BCWP = 1244800 yuan, CV = +1900 yuan (saving), SV = -28700 yuan (lagging), CPI = 1.002 ≈ 1, SPI = 0.978 < 1, determined as “cost controllable + schedule lagging”. Cost saving comes from the optimization of the material procurement plan (reducing inventory backlog).

The schedule lagging is due to the delay in the processing of the grid ceiling of the gas station canopy. The measure is to replace the supplier to shorten the processing cycle.

Table 5. Calculation table of earned value method parameters of Gas Station A project from May to July (Unit: 10000 yuan)

Month	BCWS	ACWP	BCWP	CV	SV	CPI	SPI	Management and Control Status
May	75.21	73.45	71.13	-2.33	-4.08	0.968	0.946	Cost Overrun + Schedule Lagging
June	127.35	124.48	124.48	0.19	-2.87	1.002	0.978	Cost Controllable + Schedule Lagging
July	169.46	179.26	160.87	-18.39	-8.59	0.897	0.949	Deviation Expansion

Note: BCWS = Budgeted cost of planned workload; ACWP = Actual cost of completed workload; BCWP = Budgeted cost of completed workload; CV = Cost Variance; SV = Schedule Variance; CPI = Cost Performance Index; SPI = Schedule Performance Index.

July: BCWS = 1694600 yuan, ACWP = 1792600 yuan, BCWP = 1608700 yuan, CV = -183900 yuan (serious overrun), SV = -85900 yuan (serious lagging), CPI = 0.897 < 0.9, SPI = 0.949 < 0.95, determined as “deviation expansion”. The reason is the large-scale suspension of outdoor operations caused by continuous rainfall, increasing the idle costs of labor and machinery. The measures are to add temporary rainproof facilities and supplement 2 construction teams to rush the construction of key processes.

May Deviation Response: The Cost Control Department identified the “cost overrun + schedule lagging” status and presented three options to the Project Manager: (a) maintain current resources and accept delay; (b) reallocate indoor electrical teams to accelerate pipeline work; (c) contract additional specialized crews. The Project Manager, consulting weather forecasts from the Technology Department, selected option (b), which was approved by the Owner Representative given its minimal budget impact.

4.2.4 Summary of application effects

The application of the earned value method has achieved three major transformations in the cost management and control of Gas Station A project: first, from “post-accounting” to “pre-prediction”, through the trend analysis of CPI and SPI (as shown in Figure 3), the risk of deviation expansion is predicted 1–2 months in advance; second, from “separate management and control” to “collaborative management and control”, clarifying the linkage relationship between cost overrun and schedule lagging (such as the increase of labor cost by 8000 yuan for one day of schedule lagging in July); third, from “experience-based adjustment” to “data-driven adjustment”. After the adjustment of measures, the CPI in August recovered to 0.98 and the SPI recovered to 0.97, and the cost overrun amount decreased by 62% compared with July, verifying the effectiveness of the method.

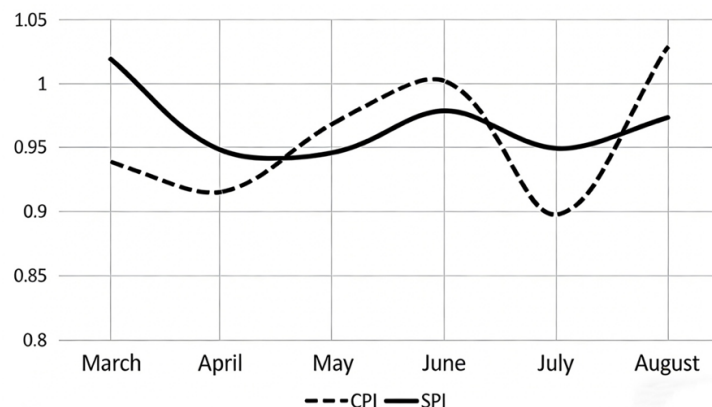


Figure 3. Trend chart of Cost Performance Index (CPI) and Schedule Performance Index (SPI) of Gas Station A project from March to August

4.3 Whole-process Optimization Strategy of Project Cost Control

4.3.1 Organizational guarantee strategy: improve the responsibility and collaboration mechanism

Based on the optimized organizational structure, further refine the cost control responsibility system: first, formulate the “Cost Control Post Responsibility Manual”, clarifying the whole-process responsibilities of the Cost

Control Department such as “plan formulation-data collection-deviation analysis-measure issuance”, as well as the cooperative responsibilities of the Engineering Department for “construction organization and cost execution” and the Finance Department for “fund accounting and budget supervision”; second, establish a weekly coordination meeting system, led by the Cost Control Department, to report deviation situations, coordinate cross-departmental issues (such as the matching degree between material procurement and construction progress), ensuring the implementation of management and control measures; third, introduce a performance appraisal mechanism, incorporate cost saving rate and schedule compliance rate into departmental KPIs, stimulating the cost management and control awareness of all employee [16].

4.3.2 Technical support strategy: promote scientific management and control tools

To address the problem of “lack of scientific management and control methods” in Gas Station A project, two technical strategies are proposed: First, expand the application scope of the earned value method, refine the management and control cycle from monthly to weekly, implement weekly deviation analysis for key projects such as the oil tank area and gas station canopy, and improve the response speed; second, explore the integration of BIM technology and cost management and control [17]. For subsequent similar projects, the visualization function of BIM models can be used to optimize construction schemes (such as optimizing the direction of gas pipelines to reduce material waste), and realize the automatic association of engineering quantity and cost data [18], improving budget accuracy. In addition, it is recommended that the project team regularly participate in cost management and control training, focusing on improving the application capabilities of the earned value method and BIM technology.

4.3.3 Plan management strategy: strengthen preliminary research and dynamic adjustment

Due to insufficient preliminary climate research, the construction period of Gas Station A project was extended, and plan management needs to be optimized from three aspects: first, improve the preliminary research system, incorporate “climate conditions, geological conditions and supply chain stability” into the necessary research content before project initiation, and formulate a “special rainy season construction plan” for rainy areas (such as the construction of temporary rainproof sheds and the staggered arrangement of indoor and outdoor processes); second, optimize the material procurement plan, combined with the characteristics of the “labor-only contract” mode, establish a linkage mechanism of “construction demand-inventory early warning-procurement execution”, with the Engineering Department submitting material demand 15 days in advance, and the Procurement Department formulating procurement plans in combination with market price fluctuations, avoiding material backlog or shortage; third, establish a dynamic plan adjustment mechanism. When the deviation between actual progress and planned progress exceeds 5%, the Cost Control Department takes the lead, jointly with the Engineering Department and Technology Department to re-evaluate the construction period and cost, and formulate adjustment plans (such as increasing construction teams and optimizing process logic).

4.3.4 Process management and control strategy: promote whole-life-cycle refinement

Taking “refinement” as the core, implement cost management and control throughout the whole life cycle of the project: First, implement “target cost design” in the early design stage, incorporate the cost ceiling of each sub-project into the design brief (such as the decoration and renovation cost of the gas station building not exceeding 153000 yuan), avoiding over-budget design schemes; second, strengthen resource management and control during the construction stage, establish an accounting system for “labor attendance-machinery use-material issuance”, calculate labor costs according to “engineering quantity $\times$ unit price”, give priority to the leasing mode for machinery use (reducing idle costs), and implement “quota issuance and surplus material recovery” for material issuance; third, strengthen final account audit during the completion stage, verify the project quantity and expense expenditure item by item against contracts and construction drawings, focus on auditing the cost of visa changes (such as the cost of temporary facilities added for rainy season construction), avoid unreasonable claims, and carry out cost post-evaluation to summarize management and control experience, providing reference for subsequent projects.

5 Conclusion

Taking the construction project of Gas Station A as the research object, this paper systematically analyzes the cost composition, existing problems and proposes optimization schemes combined with engineering project cost control theories and project actual data. The integration of machine learning extends beyond technical prediction to augment managerial cognition. By systematically processing multi-dimensional data that exceeds human analytical capacity, supporting more rational procurement decisions under supply chain uncertainty. The main research conclusions are as follows:

First, the core characteristics and key directions of gas station project cost control are clarified. As a special energy infrastructure, gas station projects have a “dual-core” cost characteristic: according to elements, material costs (33.7%) and labor costs (20.6%) account for more than 50% of the total cost; according to sub-projects, the main structure of the oil tank area (1.1739 million yuan) and the main structure of the gas station canopy (1.5848

million yuan) have the highest cost proportion, which are the core objects of cost management and control. At the same time, the project is sensitive to the external environment. For example, due to the failure to predict the local rainy season (May to September), the construction period of Project A was extended by 3 months, and the idle costs of labor and machinery increased by 438500 yuan, confirming the key role of “environmental risk prediction” in cost control.

Second, the common problems of cost control in small and medium-sized engineering projects are summarized. Through the analysis of Project A, it is found that such projects generally have four types of problems: first, the lack of an organizational system, the original structure has no special cost control department, the powers and responsibilities of each department are scattered, and cost management and control rely on the consciousness of personnel, resulting in a final project overrun of 109800 yuan; second, the one-sided consideration of the contracting mode, although the “labor-only contract” mode achieves a material cost saving of 87200 yuan, the distraction of the owner’s energy leads to an overrun of 192200 yuan in labor and machinery costs; third, insufficient preliminary research, in addition to climate, the stability of the building materials supply chain was not evaluated, and additional storage costs of 12000 yuan were incurred due to the delay in the transportation of gas station equipment; fourth, traditional management and control methods, the “post-accounting” mode led to the expansion of cost overrun from 23300 yuan to 183900 yuan from May to July, missing the opportunity for adjustment.

Third, the effectiveness of the integrated scheme of “organizational structure optimization + earned value method” is verified. At the organizational level, adding a Cost Control Department and constructing a four-level system of “Supervision Unit-Project Manager Department-Functional Departments-Cost Control Department” shortens the cost data transmission cycle from 7 days to 1 day, reducing labor cost waste by 12%; at the method level, the earned value method dynamically analyzes through three parameters: BCWS, ACWP and BCWP, identifying the overrun risk in July 2 weeks in advance. After the adjustment of measures, the CPI in August recovered from 0.897 to 0.98, and the SPI recovered from 0.949 to 0.97, with the overrun amount decreasing by 62% month-on-month, solving the pain point of “disconnection between cost and schedule” in traditional management and control.

Fourth, a whole-process management and control strategy suitable for small and medium-sized projects is formed. In the early stage, it is necessary to strengthen the three-dimensional research of “climate-geology-supply chain” and target cost design, such as limiting the decoration and renovation cost of the gas station building to within 153000 yuan; in the implementation stage, implement “quantified powers and responsibilities + dynamic adjustment”, control the material loss rate within 3%, and trigger process adjustment when the schedule deviation exceeds 5%; in the completion stage, strictly conduct final account audit and cost post-evaluation. Through audit, Project A reduced unreasonable expenditure by 21000 yuan, providing practical reference for similar projects.

This study makes three contributions to the theory of intelligent management decision-making: First, we advance understanding of DSS design [19] by demonstrating that effective intelligent tools must be co-designed with organizational structures. The Gas Station A case shows that EVM’s technical capabilities were only activated when accompanied by role specialization (Cost Control Department), information architecture (weekly reporting cycles), and decision protocols (control thresholds) [20]. This supports an organizational contingency perspective on decision-support effectiveness. Second, we develop an integrated uncertainty management framework that addresses environmental, strategic, and informational uncertainties simultaneously [21]. Prior research often examines these uncertainty types in isolation; our optimization scheme demonstrates their interactive effects and the need for multi-pronged responses.

Author Contributions

Conceptualization, X.L.X and Y.F.L.; methodology, Y.J.M.; software, Q.X.B.; validation, X.L.X, Q.X.B., and Y.F.L.; formal analysis, Y.J.M.; investigation, Y.J.M.; resources, Y.J.M.; data curation, Y.J.M.; writing—original draft preparation, Q.X.B.; writing—review and editing, X.L.X.; visualization, Y.F.L.; supervision, X.L.X.; project administration, Q.X.B.; funding acquisition, Q.X.B. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflict of interest.

References

- [1] B. Zohoori, A. Verbraeck, M. Bagherpour, and M. Khakdaman, "Monitoring production time and cost performance by combining earned value analysis and adaptive fuzzy control," *Comput. Ind. Eng.*, vol. 127, pp. 805–821, 2019. <https://doi.org/10.1016/j.cie.2018.11.019>
- [2] T. Khesal, A. Saghaei, M. Khalilzadeh, M. Rahiminezhad Galankashi, and R. Soltani, "Integrated cost, quality, risk and schedule control through earned value management (EVM)," *J. Eng. Des. Technol.*, vol. 17, no. 1, pp. 183–203, 2018. <https://doi.org/10.1108/JEDT-07-2018-0119>
- [3] Z. J. Wu, M. Y. Liu, and G. F. Ma, "A machine learning-based two-stage integrated framework for cost reasonableness prediction of green building projects," *J. Build. Eng.*, vol. 100, p. 111733, 2025. <https://doi.org/10.1016/j.jobbe.2024.111733>
- [4] T. M. O'Brien, E. Alonso, D. M. McCaffrey, and J. W. Ryter, "Multivariate regression (MVR) mining project cost estimator for future mineral supply scenario modeling," *Resour. Policy*, vol. 109, p. 105727, 2025. <https://doi.org/10.1016/j.resourpol.2025.105727>
- [5] P. Faúndez and J. I. Guzmán, "Deconstructing cost overrun and delay in mining projects: A system-based model of prices, quantities, and time," *Resour. Policy*, vol. 110, p. 105754, 2025. <https://doi.org/10.1016/j.resourpol.2025.105754>
- [6] W. J. Orlikowski, "Using technology and constituting structures: A practice lens for studying technology in organizations," *Organ. Sci.*, vol. 11, no. 4, pp. 404–428, 2000. <https://doi.org/10.1287/orsc.11.4.404.14600>
- [7] H. A. Simon, *Models of Man: Social and Rational*. New York, NY: John Wiley & Sons, 1957.
- [8] F. T. Anbari, "Earned value project management method and extensions," *Proj. Manag. J.*, vol. 34, no. 4, pp. 12–23, 2003. <https://doi.org/10.1177/875697280303400403>
- [9] D. J. Power, "Decision support systems: A historical overview," in *Handbook on Decision Support Systems*. Springer, 2008, pp. 121–140. https://doi.org/10.1007/978-3-540-48713-5_7
- [10] D. Kahneman and A. Tversky, "Prospect theory: An analysis of decision under risk," *Econometrica*, vol. 47, no. 2, pp. 263–291, 1979. <https://doi.org/10.2307/1914185>
- [11] M. S. A. Enshassi, S. Walbridge, J. S. West, and C. T. Haas, "Integrated risk management framework for tolerance-based mitigation strategy decision support in modular construction projects," *J. Manag. Eng.*, vol. 35, no. 4, p. 05019004, 2019. [https://doi.org/10.1061/\(ASCE\)ME.1943-5479.0000698](https://doi.org/10.1061/(ASCE)ME.1943-5479.0000698)
- [12] D. Jawad, Y. Medawar, and S. Ghanimeh, "Framework for conducting life cycle cost analysis for infrastructure facilities," in *ASCE India Conference 2017*, Reston, VA, 2017, pp. 450–461.
- [13] S. H. Wang, S. P. Chang, and W. C. Wang, "Experience-oriented model of budget allocation and cost control for engineering consulting projects," *Int. J. Ind. Eng.*, vol. 30, no. 2, pp. 402–420, 2023. <https://doi.org/10.2305/ijietap.2023.30.2.8421>
- [14] R. Pellegrino, "Effects of public supports for mitigating revenue risk in public-private partnership projects: Model to choose among support alternatives," *J. Constr. Eng. Manag.*, vol. 147, no. 12, p. 04021167, 2021. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0002098](https://doi.org/10.1061/(ASCE)CO.1943-7862.0002098)
- [15] M. D. Sruthi and A. Aravindan, "Performance measurement of schedule and cost analysis by using earned value management for a residential building," in *International Conference on Future Generation Functional Materials and Research (ICFMR)*, vol. 33, 2020, pp. 524–532. <https://doi.org/10.1016/j.matpr.2020.05.210>
- [16] A. M. Hamzeh and S. M. Mousavi, "A new fuzzy approach for project time assessment under uncertain conditions," in *15th Iran International Industrial Engineering Conference (IIIEC)*, Yazd, Iran, 2019, pp. 76–80. <https://doi.org/10.1109/iiiec.2019.8720733>
- [17] D. Jie and J. Wei, "Estimating construction project duration and costs upon completion using monte carlo simulations and improved earned value management," *Buildings*, vol. 12, no. 12, p. 2173, 2022. <https://doi.org/10.3390/buildings12122173>
- [18] E. Nejatyan, H. Sarvari, S. A. Hosseini, and H. Javanshir, "Determining the factors influencing construction project management performance improvement through earned value-based value engineering strategy: A delphi-based survey," *Buildings*, vol. 13, no. 8, p. 1964, 2023. <https://doi.org/10.3390/buildings13081964>
- [19] D. Przywara and A. Rak, "Monitoring of time and cost variances of schedule using simple earned value method indicators," *Appl. Sci.*, vol. 11, no. 4, p. 1357, 2021. <https://doi.org/10.3390/app11041357>
- [20] S. Babar, M. J. Thaheem, and B. Ayub, "Estimated cost at completion: Integrating risk into earned value management," *J. Constr. Eng. Manag.*, vol. 143, no. 3, p. 04016104, 2017. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001245](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001245)

- [21] Y. S. Yuan, S. D. Ye, L. Lin, and M. Gen, "Multi-objective multi-mode resource-constrained project scheduling with fuzzy activity durations in prefabricated building construction," *Comput. Ind. Eng.*, vol. 158, p. 107316, 2021. <https://doi.org/10.1016/j.cie.2021.107316>